DISCRETE STRUCTURES

Introduction to GRAPHS

What is a Graph?

Formal Definition

A graph **G** consists of two finite sets:

1. V(G): A non-empty set of vertices (points or nodes).
2. E(G): A set of edges, where each edge corresponds to a pair of vertices.

Set Representation

Example Structure:

```
V(G) = {v1, v2, v3, v4, v5}
E(G) = {e1, e2, e3, e4, e5}
```

Edges represent line segments joining pairs of vertices.

Graph Terminology

Endpoints

Vertices connected by a specific edge are called its endpoints.

Loop

An edge with just one endpoint, connecting a vertex to itself.

Adjacent

Two vertices connected by an edge are adjacent.



Incident

An edge is "incident" on each of its endpoints.

Degree of a Vertex

- **Definition:** The degree of vertex *v*, denoted *deg(v)*, is the number of edges incident on *v*.
- **The Loop Rule:** A loop contributes **2** to the degree of its vertex.
- **Isolated Vertex:** A vertex with a degree of 0 (no edges incident).

Total Degree Formula

$$\text{Sum} = \sum \deg(v) \text{ for all } v \in V(G)$$

Example Calculation:

deg(v1) = 0 (Isolated)

deg(v2) = 2

deg(v3) = 4 (Includes loop)

Total = 6

Neighbourhood of a Vertex

- **Definition:** The neighbourhood of a vertex v, denoted N(v), is a set of all vertices adjacent to v.
- **The Loop Rule:** A loop contributes **1** to the neighbourhood of its vertex.
- **Isolated Vertex:** An isolated vertex has a neighbourhood value of 0 (no edges incident).

Relation To Degree

$$\text{Deg}(v) = |N(v)|$$

Example Calculation:

N(v1) = 0 (Isolated)

N(v2) = 2

N(v3) = 4 (Includes loop)

Total = 6

Directed & Undirected Graphs

Undirected Graph



Edges consist of **unordered pairs** of vertices.

There is no specific direction associated with the edge.

(u, v) is the same as (v, u)

Directed Graph (Digraph)



Edges consist of **ordered pairs** of vertices. Each edge has a specific direction, represented by arrows.

(u, v) starts at u and goes to v

Weighted Graphs

Definition

A graph where a number (weight) is assigned to each edge. This weight typically represents cost, length, or capacity.

Applications

- **Distance:** Shortest path between cities.
- **Time:** Duration of travel.
- **Cost:** Expense of building a connection.

Isolated Graph (Null Graph)



An **Isolated Graph** (often called a Null or Empty Graph) is a graph that has vertices but **no edges**.

> $E(G) = \emptyset$ (Empty Set)

Every single vertex in this graph has a degree of 0.

Basic Types of Graphs

Simple Graph



A graph that does **not** have any loops or parallel edges.

If multiple edges connect the same vertices, it is not simple.

Complete Graph (Kn)



A simple graph where **each vertex is connected** to every other vertex.

Examples: K3 (Triangle), K4, K5.

Regular Graph

A graph **G** is regular of degree *k* (or *k*-regular) if every vertex of G has the same degree *k*.

$$\forall v \in V(G),\ \deg(v) = k$$

Exercise: *Draw two 3-regular graphs with six vertices.*

Walks, Paths & Circuits

Walk

A finite alternating sequence of adjacent vertices and edges from *v* to *w*.

Circuit

A closed walk that does not contain a repeated edge (v0 = vn).

Path

A walk that does not contain a repeated edge.

Simple Path

A path that does not contain repeated vertices.

Euler Circuits

Definition

An Euler circuit is a circuit that contains **every vertex** and **every edge** of G exactly once (edges) and at least once (vertices). It starts and ends at the same vertex.

Euler's Theorem

A graph G has an Euler circuit if, and only if:

1. G is **connected**.
2. Every vertex of G has an **even degree**.

Hamiltonian Circuits

Definition

A **Hamiltonian circuit** for graph G is a simple circuit that includes **every vertex** of G exactly once.

Unlike Euler circuits (which focus on edges), Hamiltonian circuits focus on visiting every vertex.

Planar Graphs

The Concept

A graph is called **planar** if it can be drawn in the plane without any edge crossing.

Such a drawing is called a **plane drawing** of the graph.

```
Crossing Number = 0
```

Example Logic:

Even if a graph is drawn with crossing lines initially, if it can be *redrawn* so no lines cross, it is still a planar graph.

Graph Coloring

The Problem

Assign a color to each vertex so that no two **adjacent** vertices share the same color.

Map Relation

Used to color maps where regions (vertices) sharing a border (edge) must have different colors.



Chromatic Number

The **least (minimum)** number of colors required to validly color the graph.

Questions?

Thank you for your attention.